

Saransh Aggarwal

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PROFESSIONAL SUMMARY

I am an MLOps professional specializing in the operationalization of large language models. I have hands-on experience in fine-tuning, quantizing, and deploying LLMs for real-world applications. My expertise lies in building scalable, efficient, and reliable systems for model serving and management, with a strong foundation in Python, cloud platforms (AWS), and MLOps best practices. I am passionate about bridging the gap between model development and production by creating robust and automated pipelines for LLMs.

EDUCATION

Sharda University, Greater Noida <i>B. Tech CSE - Cloud Technology (AWS Collaboration) (Fast Track)</i>	2022 – 2026 CGPA: 8.3
Lovely Public Sr. Sec. School, Delhi <i>Intermediate (CBSE)</i>	2021 – 2022 Percentage: 89.6%
Lovely Public Sr. Sec. School, Delhi <i>Matriculation (CBSE)</i>	2019 – 2020 Percentage: 80%

EXPERIENCE

Superkai <i>Product Intern</i>	October 2025 – December 2025 Delhi, India
- Architected and developed the client-server infrastructure for a Windows-based software product, ensuring low-latency communication between local clients and the backend. - Implemented secure data synchronization protocols and optimized the application logic, enhancing the overall stability and performance of the desktop environment.	

TECHNICAL SKILLS

LLM & Generative AI: LangChain, LangGraph, LLM Fine-Tuning (PEFT, LoRA), Quantization (GGUF), RAG, Vector Stores (FAISS), Prompt Engineering, Gemma Models

MLOps & Deployment: Docker, Git, GitHub, Model Serving (llama-cpp-python), API Development (Django REST Framework, Flask)

Cloud Technologies: AWS (EC2, S3, Lambda)

Programming & Data: Python, Pandas, NumPy

PROJECTS/RESEARCH

<u>Conversational Concierge for a Napa Valley Winery</u>	June 2025 – Present
- Designed and built a multi-tool conversational agent using LangGraph and a RAG pipeline with a FAISS vector store. - Developed and integrated custom tools for a real-time weather API and web search, and deployed the agent via a Streamlit UI. - Engineered a robust system prompt to guide the agent's persona, tool selection, and response synthesis.	
<u>Virtual Patient Simulator</u>	January 2025 – May 2025

- Fine-tuned a Gemma 1.1B model on a synthetically generated dataset of doctor-patient dialogues to create a persona-driven patient simulator for medical training.
- Quantized the fine-tuned model to a 4-bit GGUF format for efficient CPU-based inference and deployed it locally using Gradio and llama-cpp-python.
- Developed a Python script to generate a dataset of 1,200 training and validation examples with varied personas and dialogue structures.

Smart Shopping Assistant API

August 2024 – December 2024

- Built a real-time product information API using Django REST Framework, with a Redis caching layer to optimize for speed.
- Implemented asynchronous data fetching from Google Shopping using asyncio and aiohttp to ensure non-blocking I/O.
- Developed an interactive UI with vanilla JavaScript and deployed the application for demonstration.

CERTIFICATIONS

AWS Solution Architect Associate Certificate (AWS)

Google Cloud Computing Foundations: Cloud Fundamentals (Google)

Machine Learning Specialization (Coursera)

Web Development Bootcamp (Udemy)